

CHAPTER 5

Noah Parker

5.1. Let $f : A \rightarrow B$ be an integral homomorphism of rings. Show that $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is a closed mapping. (This is a geometric equivalent of (5.10).)

Proof. Take $\mathfrak{b} \leq B$, and recall that $f^*(V(\mathfrak{b})) \subset \overline{f^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$ by (Ex. 1.21iii). Now, suppose $\mathfrak{p} \in V(\mathfrak{b}^c)$ is a prime ideal of A , i.e. $\mathfrak{p} \supset \mathfrak{b}^c = f^{-1}(\mathfrak{b})$. Then in particular $\mathfrak{p} \supset \ker f$, so $\mathfrak{p}$ descends to a prime ideal $f(\mathfrak{p})$ of $A/\ker f \cong \text{im } f \subset B$.

$\text{im } f \subset B$ is integral, so (5.10) gives us a prime $\mathfrak{q} \in \text{Spec } B$ with $\mathfrak{q} \cap \text{im } f = f(\mathfrak{p})$. We have that $\mathfrak{q} \supset f(\mathfrak{p}) \supset f(\mathfrak{b}^c)$ implies $\mathfrak{q} \supset \mathfrak{b}^{ce} = \mathfrak{b}$, i.e. $\mathfrak{q} \in V(\mathfrak{b})$. Further,

$$\begin{aligned} f^*(\mathfrak{q}) &= f^{-1}(\mathfrak{q}) \\ &= f^{-1}(\mathfrak{q} \cap \text{im } f) \\ &= f^{-1}(f(\mathfrak{p})) \\ &= \mathfrak{p}, \end{aligned}$$

since $\mathfrak{p} \supset \ker f$. Thus, $\mathfrak{p} \in f^*(V(\mathfrak{b}))$, so in fact $f^*(V(\mathfrak{b})) = \overline{f^*(V(\mathfrak{b}))}$ is closed. $\square$

5.2. Let B be integral over $A \subset B$, and let $f : A \rightarrow \Omega$ be a homomorphism of A into an algebraically closed field Ω . Show that f can be extended to a homomorphism of B into Ω .

Proof. $f : A \rightarrow \Omega$ corresponds to a maximal ideal $\mathfrak{p} = \ker f \leq A$ and an inclusion $\bar{f} : A/\mathfrak{p} \hookrightarrow \Omega$. (5.10) supplies us with a prime ideal $\mathfrak{q} \leq B$ such that $\mathfrak{q} \cap A = \mathfrak{p}$, which is in fact maximal by (5.8). Then $\pi : B \rightarrow B/\mathfrak{q}$ restricts to a map

$$\pi|_A : A \rightarrow \frac{A}{\mathfrak{q} \cap A} = \frac{A}{\mathfrak{p}} \subset \frac{B}{\mathfrak{q}}.$$

(5.6) gives us that $K := B/\mathfrak{q}$ is integral over $k := A/\mathfrak{p}$.

Let $(k, \bar{f}) \leq (C, \bar{g})$ be a maximal element of the poset Σ of maps from subrings of K to Σ , ordered by restriction. In particular, it is a valuation ring of K , so $K \supset C \supset \bar{k} = K$ by (5.22), whence $K = C$. Then $g := \bar{g} \circ \pi : B \rightarrow \Omega$ has

$$g|_A = \bar{g}|_k \circ \pi|_A = \bar{f} \circ \pi|_A = f.$$

$\square$

5.3. Let $f : B \rightarrow B'$ be a morphism of A -algebras, and let C be an A -algebra. If f is integral, prove that $f \otimes 1 : B \otimes_A C \rightarrow B' \otimes C$ is integral.

Proof. We have that B' is integral over $f(B) \leq B'$. That is, $f(B)[b'_1, \dots, b'_n]$ is finitely generated for all $b'_1, \dots, b'_n \in B'$. Note as well that $(f \otimes 1)(B \otimes C) \cong f(B) \otimes C$. Then, for arbitrary $\sum_{i=1}^n b'_i \otimes c_i \in B' \otimes_A C$,

$$(f \otimes 1)(B \otimes_A C) \left[\sum b'_i \otimes c_i \right] \leq f(B)[b'_1, \dots, b'_n] \otimes_A C[c_1, \dots, c_n].$$

The inclusions

$$f(B) \otimes_A C \hookrightarrow f(B)[b'_1, \dots, b'_n] \otimes_A C \hookrightarrow f(B)[b'_1, \dots, b'_n] \otimes_A C[c_1, \dots, c_n]$$

are finite by closure of finiteness under base extension (2.17), so the composition is finite by closure of finiteness under composition (2.16). Then our arbitrary element of $B' \otimes_A C$ is integral over $f(B) \otimes_A C$ by (5.1iii), so that $f \otimes 1$ is integral. $\square$

5.4 Let $A \subset B$ be a subring such that B integral over A . Let $\mathfrak{n}$ be a maximal ideal of B and let $\mathfrak{m} = \mathfrak{n} \cap A$ be the corresponding maximal ideal of A . Is $B_{\mathfrak{n}}$ necessarily integral over $A_{\mathfrak{m}}$?

Proof. We will show that this need not hold in general. Consider $A := k[x^2 - 1] \subset k[x] =: B$. Clearly $k[x]$ is integral over A , as it is a finitely generated A module. Consider the maximal ideal $\mathfrak{n} = (x - 1)$. We have that $\mathfrak{n} \supset (x^2 - 1)$ and $\mathfrak{m} := (x^2 - 1)^c$ is maximal in A , so $\mathfrak{n} \cap A \supset \mathfrak{m}$ with $\mathfrak{n}$ a proper ideal implies $\mathfrak{n} \cap A = \mathfrak{m}$.

Now, suppose that $1/(x + 1) \in k[x]_{(x-1)}$ is integral over $A_{\mathfrak{m}}$. Then $A_{\mathfrak{m}}[1/(x + 1)]$ is a finitely generated $A_{\mathfrak{m}}$ module, $\square$

5.7 Let A be a subring of B such that $B - A$ is closed under multiplication. Show that A is integrally closed in B .

Proof. Suppose that $b \in B - A$ is integral over A , so that $b^{n+1} = a_n b^n + \dots + a_0$ for $a_i \in A$ and minimal $n \geq 0$. In fact, we may assume $n > 0$, as otherwise $b \in A$. Then $b^{n+1} \in B - A$ by closure under multiplication, and $a_0 \in A$, so $b^{n+1} - a_0 = a_n b^n + \dots + a_1 b \in B - A$. Suppose that

If $a_n \in A^\times$, then $b^{n-1} + a_n^{-1} a_{n-1} b^{n-1} \dots + a_n^{-1} a_1 b$ $\square$

5.10. A ring homomorphism $f : A \rightarrow B$ is said to have the *going-up property* (resp. the *going-down property*) if the conclusion of the going-up theorem (resp. going-down theorem) holds for B and its subring $f(A)$. Let $f^* : \text{Spec } B \rightarrow \text{Spec } A$ be the mapping associated with f .

(a) Consider the following three statements:

- (i) f^* is a closed mapping.
- (ii) f has the going-up property.
- (iii) For any prime ideal $\mathfrak{q}$ of B , let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$ is surjective.

Prove that (i) $\implies$ (ii) $\iff$ (iii).

(b) Consider the following three statements:

- (i) f^* is an open mapping.
- (ii) f has the going-down property.
- (iii) For any prime ideal $\mathfrak{q}$ of B , let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \text{Spec}(B_{\mathfrak{q}}) \rightarrow \text{Spec}(A_{\mathfrak{p}})$ is surjective.

Prove that (i) $\implies$ (ii) $\iff$ (iii).

Proof. (a) We begin by showing that (i) $\implies$ (ii). $\square$

5.16. Let k be a field and let $A \neq 0$ be a finitely generated k -algebra. Then there exist elements $y_1, \dots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \dots, y_r]$.

We shall assume that k is *infinite*. (A different proof is required for the finite case.) From the proof, it follows that $y_1, \dots, y_r$ may be chosen to be linear combinations of $x_1, \dots, x_n$. This has the following geometrical interpretation: if k is algebraically closed and X is an affine algebraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L . [Use Exercise (2).]

- 5.27 Let A, B be two local rings. B is said to *dominate* A if A is a subring of B and the maximal ideal $\mathfrak{m}$ of A is contained in the maximal ideal $\mathfrak{n}$ of B (or, equivalently, if $\mathfrak{m} = \mathfrak{n} \cap A$).

Let K be a field and let Σ be the set of all local subrings of K . If Σ is ordered by the relation of domination, show that Σ has maximal elements, and that $A \in \Sigma$ is maximal if and only if A is a valuation ring of K .

Proof. Consider the poset

$$\Sigma' = \{(A, f) \mid A \leq K, f : A \rightarrow \overline{K}\}$$

ordered by restriction. Clearly $\Sigma \subset \Sigma'$ as sets by identifying $A \leq K$ with $(A, A \hookrightarrow K \hookrightarrow \overline{K})$. It is clear that $A \leq_{\Sigma} B$ implies $(A, i : A \hookrightarrow \overline{K}) \leq_{\Sigma'} (B, j : B \hookrightarrow \overline{K})$. Conversely, suppose $(A, i) \leq_{\Sigma'} (B, j)$. Then A is a subring of B , and

$$\mathfrak{m} = \ker i = \ker j|_A = \ker j \cap A = \mathfrak{n} \cap A$$

by (5.19), so that $A \leq_{\Sigma} B$. Thus, their poset structures coincide.

Suppose that $A \in \Sigma$ is maximal. If $(A, i) \leq_{\Sigma'} (B, g)$, then $\square$

- 5.28 Let A be an integral domain, K its field of fractions. Show that the following are equivalent:

- (a) A is a valuation ring of K ;
- (b) If $\mathfrak{a}, \mathfrak{b}$ are any two ideals of A , then either $\mathfrak{a} \subset \mathfrak{b}$ or $\mathfrak{b} \subset \mathfrak{a}$.

Deduce that if A is a valuation ring and $\mathfrak{p}$ is a prime ideal of A , then $A_{\mathfrak{p}}$ and $A/\mathfrak{p}$ are valuation rings of their fields of fractions.

Proof. $\square$

- 5.29 Let A be a valuation ring of a field K . Show that every subring of K which contains A is a local ring of A .

Proof. $\square$

- 5.30 Let A be a valuation ring of a field K . The group $A^{\times}$ of units of A is a subgroup of the multiplicative group $K^{\times}$ of K .

Let $\Gamma = K^{\times}/A^{\times}$. If $\xi, \eta \in \Gamma$ are represented by $x, y \in K$, define $\xi \geq \eta$ to mean $xy^{-1} \in A$. Show that this defines a total ordering on Γ which is compatible with the group structure (i.e., $\xi \geq \eta \implies \xi\omega \geq \eta\omega$ for all $\omega \in \Gamma$). In other words, Γ is a totally ordered abelian group. It is called the *value group* of A .

Let $v : K^{\times} \rightarrow \Gamma$ be the canonical homomorphism. Show that $v(x + y) \geq \min(v(x), v(y))$ for all $x, y \in K^{\times}$.

Proof. $\square$

- 5.31 Conversely, let Γ be a totally ordered abelian group (written *additively*), and let K be a field. A *valuation of K with values in Γ* is a mapping $v : K^{\times} \rightarrow \Gamma$ such that

(a) $v(xy) = v(x) + v(y)$,
 (b) $v(x + y) \geq \min(v(x), v(y))$,
 for all $x, y \in K^\times$. Show that the set of elements $x \in K$ such that $x = 0$ or $v(x) \geq 0$ is a valuation ring of K . This ring is called the valuation ring of v , and the subgroup $v(K^\times)$ is the *value group* of v .

Thus, the concepts of valuation ring and valuation are essentially equivalent.

Proof.

□

5.35 From (Ex. 5.1) and (Ex. 5.3), it follows that, if $f : A \rightarrow B$ is integral and C is any A -algebra, then the mapping $(f \otimes 1)^* : \text{Spec}(B \otimes_A C) \rightarrow \text{Spec } C$ is a closed map.

Conversely, suppose that $f : A \rightarrow B$ has this property and that B is an integral domain. Then f is integral.

Show that the result just proved remains valid if B is a ring with only finitely many minimal prime ideals (e.g., if B is Noetherian).

Proof.

□